American Computer Science League

2024-2025 ● Contest 2: Short Problems ● Classroom Division

1. Prefix/Infix/Postfix Notation

Convert the following infix expression to prefix:

3 + 4 * (8 − 4 / 2)

A. + 3 4 * − 8 / 4 2
B. + 3 * 4 − 8 / 4 2
C. + 3 * 4 − / 8 4 2
D. + 3 * − 4 8 / 4 2
E. + * 3 4 − 8 / 4 2

2. Prefix/Infix/Postfix Notation

Evaluate the following postfix expression if all numbers are single digits:

9 8 6 2 − * 2 / + 6 3 / −

A. 8
B. 14
C. 20
D. 23
E. 27

3. Bit-String Flicking

Evaluate:

01011 OR NOT 01101

A. 11011
B. 01011
C. 10010
D. 01101
E. 11001

4. Bit-String Flicking

Evaluate:

(LSHIFT-2 01101) OR (RCIRC-3 10110) AND (RSHIFT-1 11011)

A. 01000
B. 10100
C. 11100
D. 01101
E. 11010

5. Prefix/Infix/Postfix Notation

Evaluate the following prefix expression if all numbers are single digits:

+ − * 2 3 / 4 2 / * 8 + 1 3 4

A. 4
B. 6
C. 8
D. 10
E. 12

6. Prefix/Infix/Postfix Notation

Translate the following infix expression into postfix:

$$\frac{xy + z}{a^2}$$

A. x y z * + a 2 ^ /
B. x y * + z a 2 ^ /
C. x y * z + a 2 ^ /
D. x * y z + a 2 ^ /
E. x y * z a 2 ^ + /

7. Bit-String Flicking

Evaluate:

10110 OR 11011 AND 00101 XOR 01101

A. 11110
B. 00011
C. 01110
D. 11011
E. 01000

8. Bit-String Flicking

Solve the following equation for X if X is a 5-bit string:

(NOT (RCIRC-3 (LSHIFT-2 X))) = 01100

A. **101
B. 1*110
C. *1000
D. **111
E. 0*011

9. LISP

Evaluate the following LISP expression:

(ADD 2 (SUB 7 3) 3 (MULT 2 5) 8)

A. 11
B. 13
C. 23
D. 25
E. 27

10. LISP

Evaluate the following LISP expression:

(CDR (CAR (CDR (CDR '(e (f g) (d b (a c)))))))

A. (d b (a c))
B. (b (a c))
C. (f g)
D. (b)
E. (e (f g))